Homework1

January 30, 2025

1 1.

[1]:
```
def fibonacci(n):
    x = 0
    y = 1

    numbers = []

    for i in range(n):
        numbers.append(x)
        z = y
        y = x + z
        x = z

    return numbers

fibonacci(20)
```

[1]:
```
[0,
 1,
 1,
 2,
 3,
 5,
 8,
 13,
 21,
 34,
 55,
 89,
 144,
 233,
 377,
 610,
 987,
 1597,
 2584,
```

```
 4181]
```

2 2.

[2]:
```python
import math

def IsPrime(n):
    b = True
    for i in range(2,int(math.sqrt(n))+1):
        if n % i == 0:
            b = False

    return b

print(IsPrime(4))
print(IsPrime(23))
```

```
False
True
```

[3]:
```python
def primes(n):
    primes = []
    for i in range(2,n+1):
        if IsPrime(i):
            primes.append(i)

    return primes

primes(100)
```

[3]:
```
[2,
 3,
 5,
 7,
 11,
 13,
 17,
 19,
 23,
 29,
 31,
 37,
 41,
 43,
 47,
 53,
 59,
```

```
61,
67,
71,
73,
79,
83,
89,
97]
```

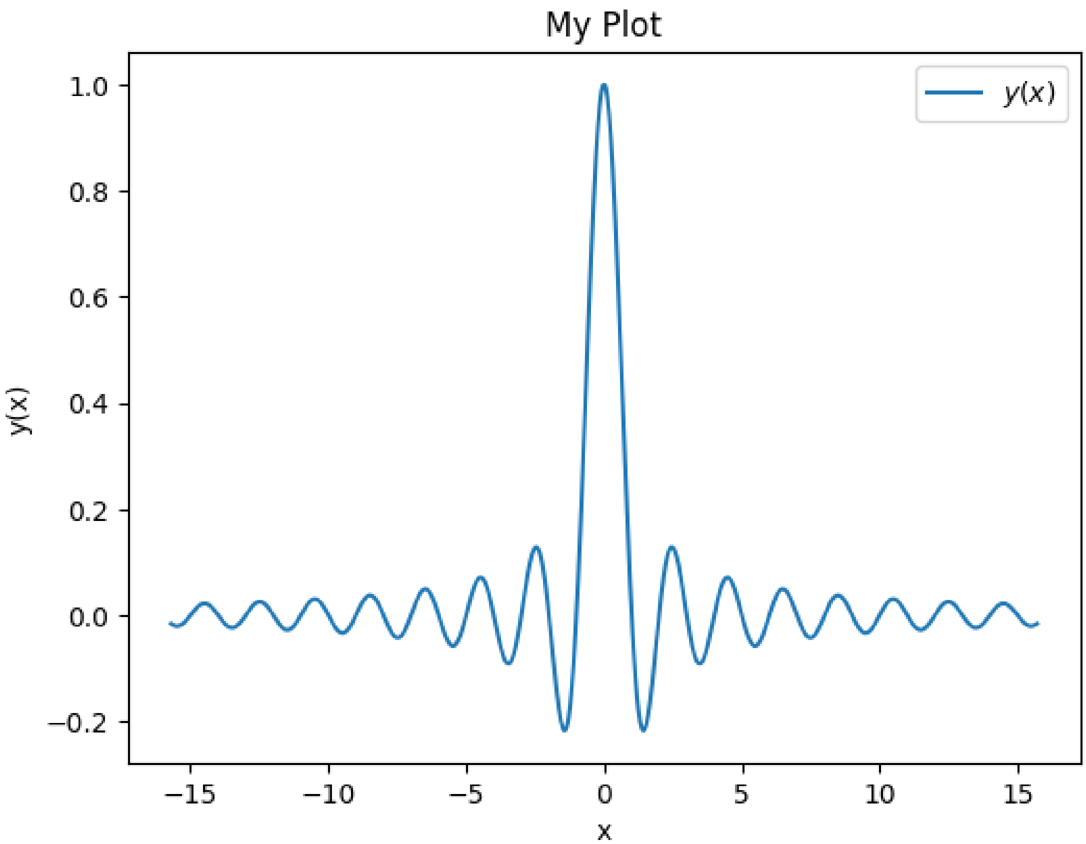
3 3.

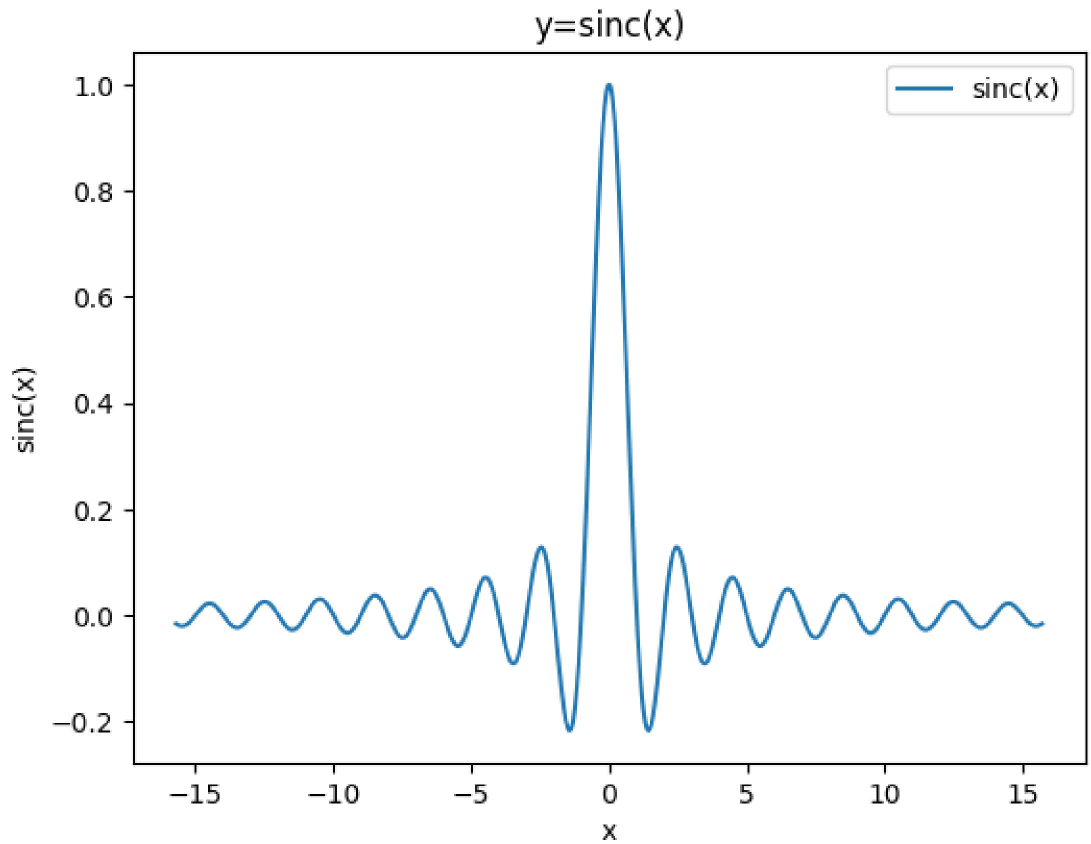
[4]:
```
import numpy as np
import matplotlib.pyplot as plt

def MyPlot(x, y, xlabel='x', ylabel='y(x)', title='My Plot', label='$y(x)$'):
    plt.plot(x,y,label=label)
    plt.title(title)
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.legend()
    plt.show()

x_vals = np.linspace(-5*np.pi, 5*np.pi, 1001)
y_vals = np.sinc(x_vals)

MyPlot(x_vals,y_vals)
MyPlot(x_vals,y_vals,ylabel='sinc(x)',title='y=sinc(x)',label='sinc(x)')
```


My Plot
y(x)
$y(x)$
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−15
−10
−5
0
5
10
15
x



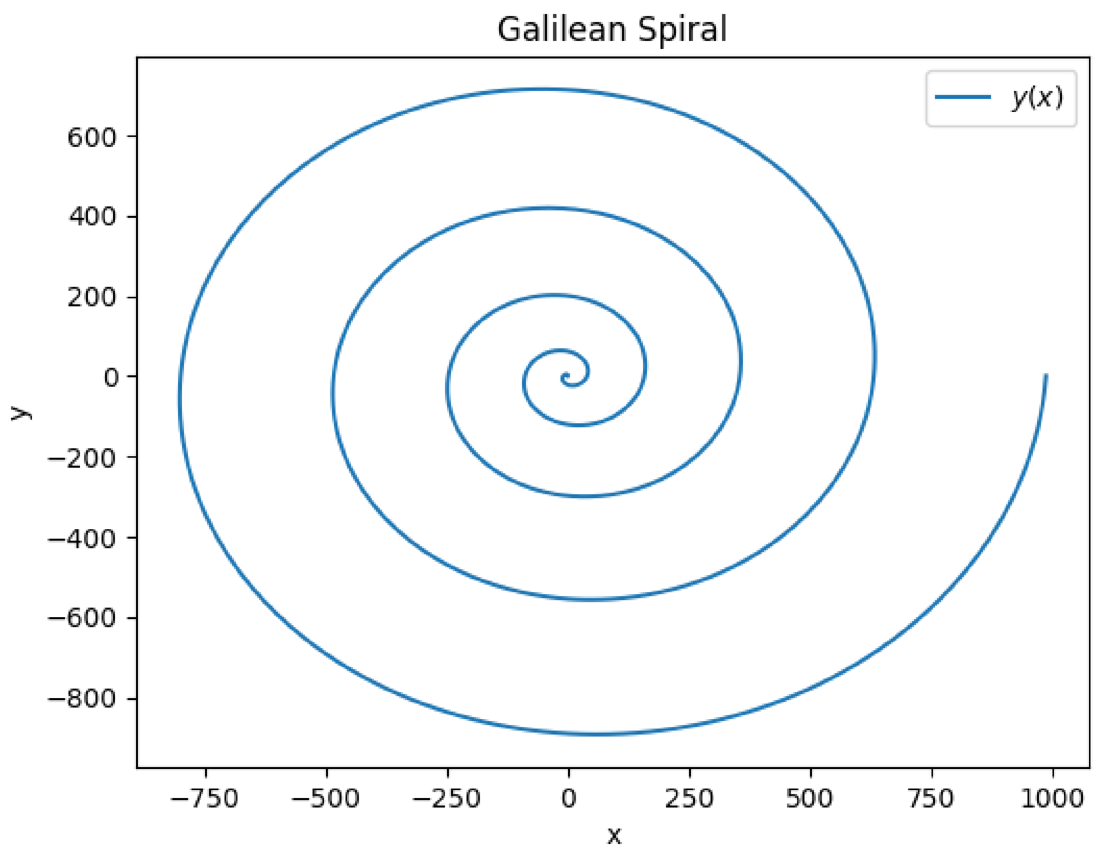
4 4.

4.1 a)

[5]:
```
theta = np.linspace(0, 10*np.pi, 1001)
r = theta ** 2

x = r * np.cos(theta)
y = r * np.sin(theta)

MyPlot(x,y,title="Galilean Spiral",ylabel='y')
```

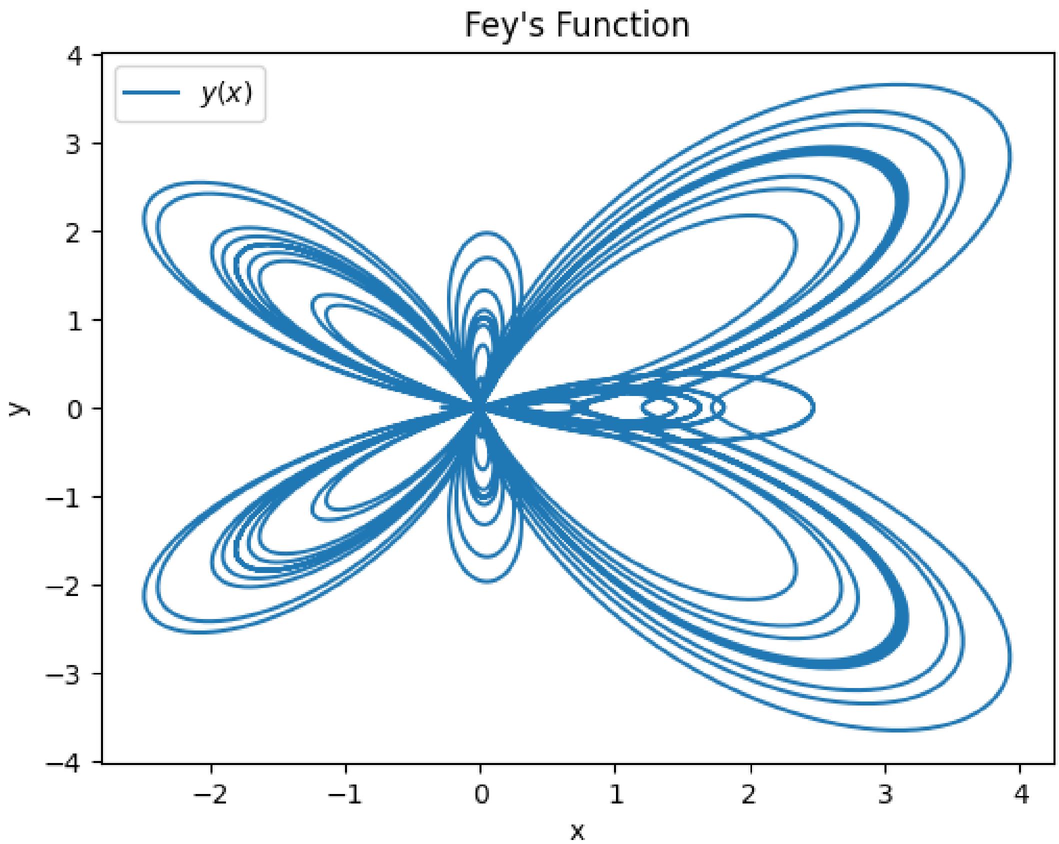


4.2 b)

[6]:
```
theta = np.linspace(0, 24*np.pi, 5001)
r = np.exp(np.cos(theta)) - 2*np.cos(4*theta) + (np.sin(theta/12))**5

x = r * np.cos(theta)
y = r * np.sin(theta)

MyPlot(x,y,title="Fey's Function",ylabel="y")
```


Fey's Function
y(x)
y
x
4
3
2
1
0
−1
−2
−3
−4
−2
−1
0
1
2
3
4